MILITARY SPECIFICATION

BOOSTER, FUZE, MARK 38 MOD 2

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 38 MOD 2 Fuze Booster, referred to herein as the booster.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/8	Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 6061.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

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MILITARY

MIL-A-2550	Ammunition, General Specification for.
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for).
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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OTHER GOVERNMENT DOCUMENTS

DEPARTMENT OF DEFENSE (DoD)

DoD 4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND
(Code Ident 10001)

LD 521855	Fuze Booster, MARK 38 MOD 1 (Special Packaging) (For 175 MARK 38 MOD 1 Fuze Boosters).

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

SA 2618609	Fixture, Drop Test, Safety-Arming Device, MARK 5 All MODs.
SA 917AS3145	Safety-Arming Device, Dummy.
917AS3163	Booster, Fuze, MARK 38 MOD 2.

(Copies of drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

PUBLICATIONS

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A-108	Steel Bars, Carbon, Cold-finished, Standard Quality, Specification for.
ASTM A-519	Seamless Carbon and Alloy Steel Mechanical Tubing, Specification for.

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ASTM A-581	Free-machining Stainless and Heat-resisting Steel Wire, Specification for.
ASTM A-582	Free-machining Stainless and Heat-resisting Steel Bars, Hot-rolled or Cold-finished, Specification for.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadephia, PA 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description. The booster initiates the warhead of a guided missile and is composed of the components shown on Figure 1. The booster pellet contains approximately 44 grams of Composition CH-6 explosive.

3.1.1 Government-furnished property. Unless otherwise specified in the contract or purchase order, the Government will furnish the contractor with the following in the quantity specified (see 6.2.3):

a. Composition CH-6 explosive (MIL-C-21723); approximately 120 percent of the net production requirement.

b. Explosive Lead, MARK 8 MOD 0 (LD 480140); approximately 120 percent of the total test quantity requirement.

c. Explosive Lead, MARK 16 MOD 0 (LD 521882); approximately 120 percent of the total test quantity requirement.

d. Exploding bridgewire (EBW) detonator (see 4.6.1.1.2.a); approximately 120 percent of the total test quantity requirement.

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3.2 Performance characteristics.

3.2.1 Function and output. The booster shall function when initiated in accordance with 4.6.1 and the output shall meet the acceptance criteria specified in 4.6.1.3.

3.2.2 Environmental conditions. The booster shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the following environments.

3.2.2.1 Transportation vibration-temperature. After exposure to the environment of 4.6.2.1, the booster shall be safe (see 6.3) and, when initiated in accordance with 4.6.1, shall function and produce the proper output (see 3.2.1).

3.2.2.2 Temperature and humidity. After exposure to the environment of 4.6.2.2, the booster shall be safe (see 6.3) and, when initiated in accordance with 4.6.1, shall function and produce the proper output (see 3.2.1).

3.2.2.3 Captive carry and missile flight vibration. After exposure to the environment of 4.6.2.3, the booster shall be safe (see 6.3) and, when initiated in accordance with 4.6.1, shall function and produce the proper output (see 3.2.1).

3.2.3 Environmental safety.

3.2.3.1 Forty-foot drop. During exposure to the environment of 4.6.3.1, the booster shall not explode nor burn. After exposure, the booster shall be safe (see 6.3) to handle and dispose of.

3.2.3.2 Jolt. After exposure to the environment of 4.6.3.2, the booster shall be safe (see 6.3) to store, handle, and transport.

3.2.3.3 Cook-off. During and after exposure to the environment of 4.6.3.3, the booster shall not detonate. Any booster reaction (deflagration, burning, etc.), other than detonation, is acceptable and may occur at any time.

3.3 Mechanical characteristics.

3.3.1 Radiographic. When radiographically inspected, the booster shall show no internal evidence of cracks, fissures, loose powder, or any foreign material, and all components and parts shall be properly located as shown on Drawing 917AS3163 (see 4.6.4).

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3.3.2 Visual. When visually inspected, the booster shall show no external evidence of cracks, burrs, dents, loose powder, or other deformities which may interfere with functioning or fit at the next assembly (see 4.6.5).

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2.4).

3.5 Design and construction.

3.5.1 Production drawings. The booster shall meet the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on Drawing 917AS3163.

3.5.2 Standards of manufacture.

3.5.2.1 Safety precautions. As a minimum, the safety precautions of DoD 4145.26M shall apply during manufacture, handling, and inspection of the booster, but shall not be considered an adequate substitution for the contractor's own safety procedures developed to accommodate the hazards associated with the booster.

3.5.2.2 Working environment. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.5.2.3 Test plan and test procedures. When specified in the contract or purchase order, a test plan and test procedures shall be prepared by the contractor in accordance with MIL-T-18303 and the Data Item Description (DID) cited in 6.2.2. The test plan and test procedures shall be accepted by the Government prior to use.

3.5.2.4 Inspection record. When specified in the contract or purchase order, an inspection record shall be prepared and filled in by the contractor in accordance with the DID cited in 6.2.2 for each booster produced under a specified contract. The required test data shall be extracted from the inspection and acceptance records maintained by the contractor.

3.5.2.5 Workmanship. Workmanship during manufacture of the booster shall meet the requirements of MIL-A-2550.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.4).

b. Quality conformance inspections (see 4.5).

4.2 Test equipment. All test equipment utilized by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662.

4.2.1 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements, under the conditions and to the accuracies specified herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured. All test equipment shall be calibrated and maintained as specified in 4.2.

4.2.2 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein, are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.2.1 and are subject to approval by the procuring activity.

4.2.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. When specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

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4.3 Test conditions. Unless otherwise specified herein, the inspection methods of 4.6 shall be performed under the conditions specified in 3.5.2.2.

4.4 First article inspections. When specified in the contract or purchase order, first article inspections shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government (see 6.2.4). First article inspections shall be performed on 30 sample boosters which have been produced with equipment and procedures to be used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

4.4.1 First article inspection sequence. The first article of 30 boosters shall be subjected to the inspections of Figure 2 in the sequence shown. After tests, the boosters shall be disposed of at the discretion of the cognizant demolition officer.

4.4.2 First article acceptance and rejection criteria. Each first article booster shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 2, the applicable performance requirements of Section 3 shall be met on the first and only trial; one or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspections.

4.5.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot shall contain not less than 500 nor more than 1,000 boosters (see 6.2.1). An inspection lot shall consist of completed boosters fabricated under the same conditions and offered for acceptance inspection at the same time.

4.5.2 Inspection lot inspection sequence. After the inspection lot has been formed and identified by lot number, the lot shall be subjected to the inspections of Figure 3 in the sequence shown. Where specified during the inspection sequence, a quantity of boosters shall be randomly sampled from the inspection lot in accordance with MIL-STD-414, Inspection Level IV, Standard Deviation Method, Variability Unknown. While the sample continues through the inspection sequence, the remainder of the lot shall be held in bond pending acceptance of the sample. If the sample does not cause lot rejection, the remainder of the lot shall continue through the inspection sequence.

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4.5.3 Inspection lot acceptance and rejection criteria. For the inspections of Figure 3, inspection lot acceptance and rejection shall be based on the following criteria:

a. Where 100 percent inspection is required (4.6.4, 4.6.5, and 4.6.6), acceptance or rejection shall be on an individual basis. The failure of any booster to meet the requirements specified in the contract or purchase order, on the drawings, or herein shall cause rejection of the booster.

b. Where sampling is required, lot acceptability for the function and output test of 4.6.1 shall be in accordance with 4.6.1.3.

4.6 Inspection methods. To determine conformance to the applicable requirements of Sections 3 and 5, the following inspections shall be performed in accordance with the inspection sequences of Figures 2 and 3, as applicable.

WARNING

The booster contains explosive which is highly sensitive and hazardous. Suitable safety precautions and procedures must be observed during handling and performance of the inspections specified herein (see 3.5.2.1).

4.6.1 Function and output. The function and output test consists of initiating the booster on a steel dent block (see 4.6.1.1.1.e) and measuring the depth of the dent to determine the output. The test operation and selection of test equipment and materials, except for those listed in 4.6.1.1.1, shall be the responsibility of the contractor. A list of recommended test equipment and materials is given in 4.6.1.1.2 to facilitate test setup by the contractor. To determine conformance to 3.2.1, test the booster in accordance with 4.6.1.2; then measure the depth of the dent and perform the computation as specified in 4.6.1.3.

4.6.1.1 Function and output test equipment and materials.

4.6.1.1.1 Required test equipment and materials. The following test equipment and materials are required to perform the function and output test:

a. Lead holder, spacer, and aluminum barrier in accordance with Figures 4a, 4b, and 4c, respectively.

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b. Explosive Lead, MARK 8 MOD 0 (see 3.1.1).

c. Explosive Lead, MARK 16 MOD 0 (see 3.1.1).

d. Detonator (an electric EBW detonator is recommended; see 3.1.1 and 4.6.1.1.2.a).

NOTE: Unless otherwise specified in the contract or purchase order, the Government will supply the contractor with EBW detonators (see 3.1.1). A contractor-selected detonator may be used subject to the approval of the output characteristics by the procuring activity. Approval of the contractor-selected detonator does not imply approval of the detonator's safety characteristics.

e. Dent block as specified in MIL-STD-331, Test 301, except the size of the block shall be 5.000 ±0.125 inches square and shall be cut from 1.25 inch stock.

f. Expendable support blocks to sustain the dent block of e. above (material and size shall be at the discretion of the contractor).

g. Dial indicator height gage and probe as specified in MIL-STD-331, Test 301.

h. Detonator firing source, low current resistance measurement device with the current limited to 10 milliamperes (mA) maximum, and interconnecting cables for the detonator of d. above.

4.6.1.1.2 Recommended test equipment and materials. The following test equipment and materials are recommended for safe performance of the function and output test in conjunction with suitable safety procedures and facilities:

a. EBW detonator of 0.190 ±0.001 inch diameter with an output charge of 70 ±20 milligrams of pentaerythritol tetranitrate (PETN) pressed at 10,000 ±500 pounds per square inch (lb/in^2) (see 3.1.1).

b. Firing source and cables for the detonator of a. above.

c. Firing bay or barricades for the blast and fragment hazards of the booster.

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4.6.1.2 Function and output test method. Test the booster as specified below using the test setup shown on Figure 5:

WARNING

The function and output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 3.5.2.1).

a. Place the steel dent block on appropriate expendable support blocks as shown on Figure 5.

b. Place the booster in the center of the dent block, booster cap (see Drawing 917AS3144) up.

c. Place the aluminum barrier (see Figure 4c) on the booster cap.

d. Place the loaded lead holder (see Figures 4a and 5), without the detonator, on the aluminum barrier.

NOTE 1: The installation and wiring of the detonator shall be performed in accordance with the contractor's safety procedures.

NOTE 2: Steps e. through m. below are suggested as good safety procedures appropriate for the hazards involved.

e. Clear the firing area of all nonessential personnel and equipment.

f. Verify that the detonator leads are shorted and that ordnance personnel, the firing cables, and the booster assembly are grounded.

g. Verify that the firing cables are not connected to the firing source.

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h. Insert the detonator into a hand-safe containment block, remove the short from the detonator leads, connect the detonator leads to the firing cable, and insulate the connections.

i. Insert the detonator into the lead holder (see Figures 4a and 5) so the detonator is bottomed against the shoulder, as shown on Figure 5.

j. Remove all remaining personnel and nonexpendable equipment from the test bay and verify that all openings through which fragments could escape are secured.

WARNING

The booster causes a significant blast and fragment hazard. Verify that the area is cleared of all personnel before applying energy to the firing circuit (see 3.5.2.1).

k. Connect the firing cable to the firing source.

l. Verify that all personnel are clear, sound the firing alarm, and initiate the detonator.

m. Allow sufficient time to elapse for the fumes to clear the firing bay, resecure the firing source, disconnect and short the firing cables, and retrieve the dent block.

4.6.1.3 Output dent measurement procedure and acceptance critieria. Measure the depth of dent as follows to determine conformance to the acceptance criteria specified below:

a. Measure the depth of dent as specified in MIL-STD-331, Test 301. A depth of dent of less than 0.125 inch from any one booster shall cause rejection of the first article or the inspection lot.

b. Using the data from a. above, compute the quantity $(\bar{X}-L)/s$, as defined in MIL-STD-414, Single Specification Limit, Form 1, Variability unknown, Standard Deviation Method. The lower limit, L, of the booster output to be used in the computation shall be 0.130 inch. The first article or the inspection lot shall be accepted if the quantity $(\bar{X}-L)/s$ is equal to or greater than MIL-STD-414, Table B-1, k value, acceptable quality level (AQL) of 0.65 percent defective. A quantity less than the k value shall cause rejection of the first article or the inspection lot.

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4.6.2 Environmental.

4.6.2.1 Transportation vibration-temperature. To determine conformance to 3.2.2.1, install the booster in the shipping and storage container (see LD 521855) and subject the booster to the test of MIL-STD-331, Test 119, Procedure 1, except:

a. The control accelerometer shall be mounted on the shaker table as close as possible to the shipping and storage container.

b. The vibration test duration shall be 2 hours minimum per axis.

c. For the first article (see Figure 2), the temperature conditions shall be -40 ±5 and +140 ±5 degrees Fahrenheit (°F).

4.6.2.2 Temperature and humidity. To determine conformance to 3.2.2.2, subject the booster to the test of MIL-STD-331, Test 105, either method, except the test shall be conducted for one 14-day period. Thoroughly dry the booster after the test.

4.6.2.3 Captive carry and missile flight vibration. To determine conformance to 3.2.2.3, subject the booster to the test of MIL-STD-810, Method 514, Equipment Category d.1, Procedure IIA, except:

a. A vibration test adapter shall be securely mounted to a block-type fixture, which may be configured to hold more than one booster and simultaneously test boosters in more than one axis, providing the other conditions of the test are satisfied.

NOTE: Unless otherwise specified in the contract or purchase order, the vibration test adapter of a. above shall be approved by the procuring activity prior to use (see 6.2.1).

b. To provide corroboration of the input accelerations delivered, a minimum of one booster per axis shall be randomly selected and affixed with a response accelerometer.

c. Excitation in accordance with the curve of Figure 6 shall be applied separately along each of the three orthogonal axes of the booster for 2 hours minimum per axis (for a total vibration time of 6 hours minimum per booster).

d. For the first article (see Figure 2), 10 boosters shall be preconditioned for 4 hours minimum at -65 ±5°F and vibrated at this temperature; and 10 boosters shall be preconditioned for

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4 hours minimum at +165 ±5°F and vibrated at this temperature. Between axes (see c. above), the boosters shall be reconditioned to the required temperature for twice the time out of temperature or for 4 hours minimum, whichever is less.

4.6.3 Environmental safety.

4.6.3.1 Forty-foot drop. To determine conformance to 3.2.3.1, test the booster as follows:

a. Attach the booster to the dummy safety-arming device of Drawing SA 917AS3145.

b. Install the dummy safety-arming device with the booster attached into the fixture of Drawing SA 2618609, and torque the mounting serews to 24 ±2 pound-inches.

c. Subject the assembly of b. above to the test of MIL-STD-331, Test 103, Procedure 1, except the warhead test fixture shall be dropped once in each of the following orientations: horizontal, nose up, and nose down.

d. After test, examine the booster to determine conformance to 3.2.3.1. Breakage of the explosive pellet, the plastic cup, or the presence of loose pieces of explosive in the warhead fixture central tube shall not be considered criteria for failing the test.

4.6.3.2 Jolt. To determine conformance to 3.2.3.2, subject the booster to the test of MIL-STD-331, Test 101, Procedure II.

4.6.3.3 Cook-off. To determine conformance to 3.2.3.3, test the booster as follows:

WARNING

The cook-off test involves hazardous operations. Suitable safety precautions must be observed (see 3.5.2.1).

a. Install the booster in the cook-off test fixture of Figure 7 as shown on Figure 8.

NOTE: Use of a high temperature anti-seize compound between all threaded surfaces will greatly facilitate disassembly after the test.

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b. Position the assembly of a. above over a gas burner array as shown on Figure 9. The height of the assembly above the array shall be such that the assembly will be engulfed in flame.

NOTE: Unless otherwise specified in the contract or purchase order, the test setup of b. above shall be approved by the procuring activity prior to test (see 6.2.1).

WARNING

During the direct flame impingement of c. below, the simulated warhead central tube (see Figure 7) may fail in an overpressure/rupture mode. Some portions of the simulated warhead central tube may be expelled violently.

c. Subject the assembly to direct flame impingement at an average flame temperature of 1,500°F until all ordnance reactions are completed or until 15 minutes maximum have elapsed. Continuously monitor and record the temperature of the flame at the two locations indicated on Figure 9, and the temperature of the simulated warhead central tube.

4.6.4 Radiographic. To determine conformance to 3.3.1, perform a radiographic inspection of each booster in accordance with MIL-STD-453, Quality Level 2. Take a minimum of two X-rays of each booster, each 90 ±5 degrees apart and showing the longitudinal section through the booster (see Figure 1).

4.6.5 Visual. To determine conformance to 3.3.2 and the general requirements of Section 3, visually inspect the booster.

4.6.6 Drawing conformance. To determine conformance to 3.5.1, perform drawing conformance inspections on a sample lot of parts and subassemblies of the booster during manufacture, and on each completely assembled booster in accordance with the drawings listed on Drawing 917AS3163. Sampling and acceptance criteria shall be as specified on the drawings.

4.6.7 Packaging. To determine conformance to Section 5, inspect the packaging, packing, and marking of the booster prior to shipment.

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5. PACKAGING.

5.1 Preservation - packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and the drawings and documents listed on LD 521855.

5.1.1 Level C packaging. Boosters shall be packaged in commercially-available materials suitable for explosives and in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity. Packaging shall conform to 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and the drawings and documents listed on LD 521855.

5.2.1 Level C packing. Boosters, packaged in accordance with 5.1.1, shall be packed in commercially-available materials suitable for explosives and in a manner that will provide protection from damage during direct shipment from the supply source to the first receiving activity. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, marking shall be in accordance with MIL-STD-129 and the drawings and documents listed on LD 521855 (see 6.2.1).

6. NOTES.

6.1 Intended use. The booster is intended for use in the warhead explosive train of the AIM-7F and AIM/RIM-7M Sparrow Guided Missiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

c. Government final inspection gage drawings, if required (see 4.2.3).

d. Inspection lot size, if other than as specified in 4.5.1.

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e. Approval by procuring activity of the test fixture for the captive carry and missile flight vioration test, if required (see Note, 4.6.2.3).

f. Approval by procuring activity of cook-off test setup, if required (see Note, 4.6.3.3).

g. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulations (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirements	Applicable DID	Options
3.5.2.3	Test Plan and Test Procedures	DI-T-5204	---
3.5.2.4	Inspection Record	DI-E-2001	---

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

6.2.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 30 boosters, as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the document's first article.

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6.3 Definition of safe. Safe is defined as the ability of the booster to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the booster to maintain its safety features in a condition which will permit disposal without injury to personnel.

Preparing activity:
Navy - AS
(Project 1336-N391)

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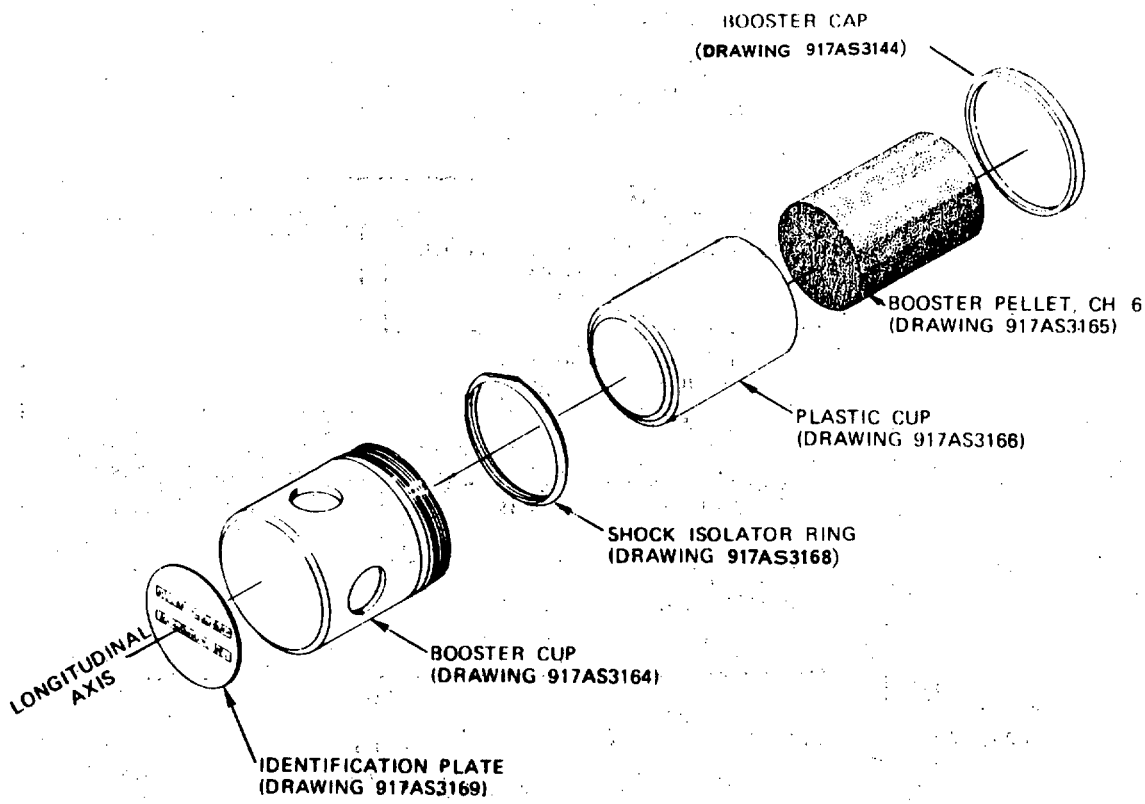


FIGURE 1. MARK 38 MOD 2 Fuze Booster major components.

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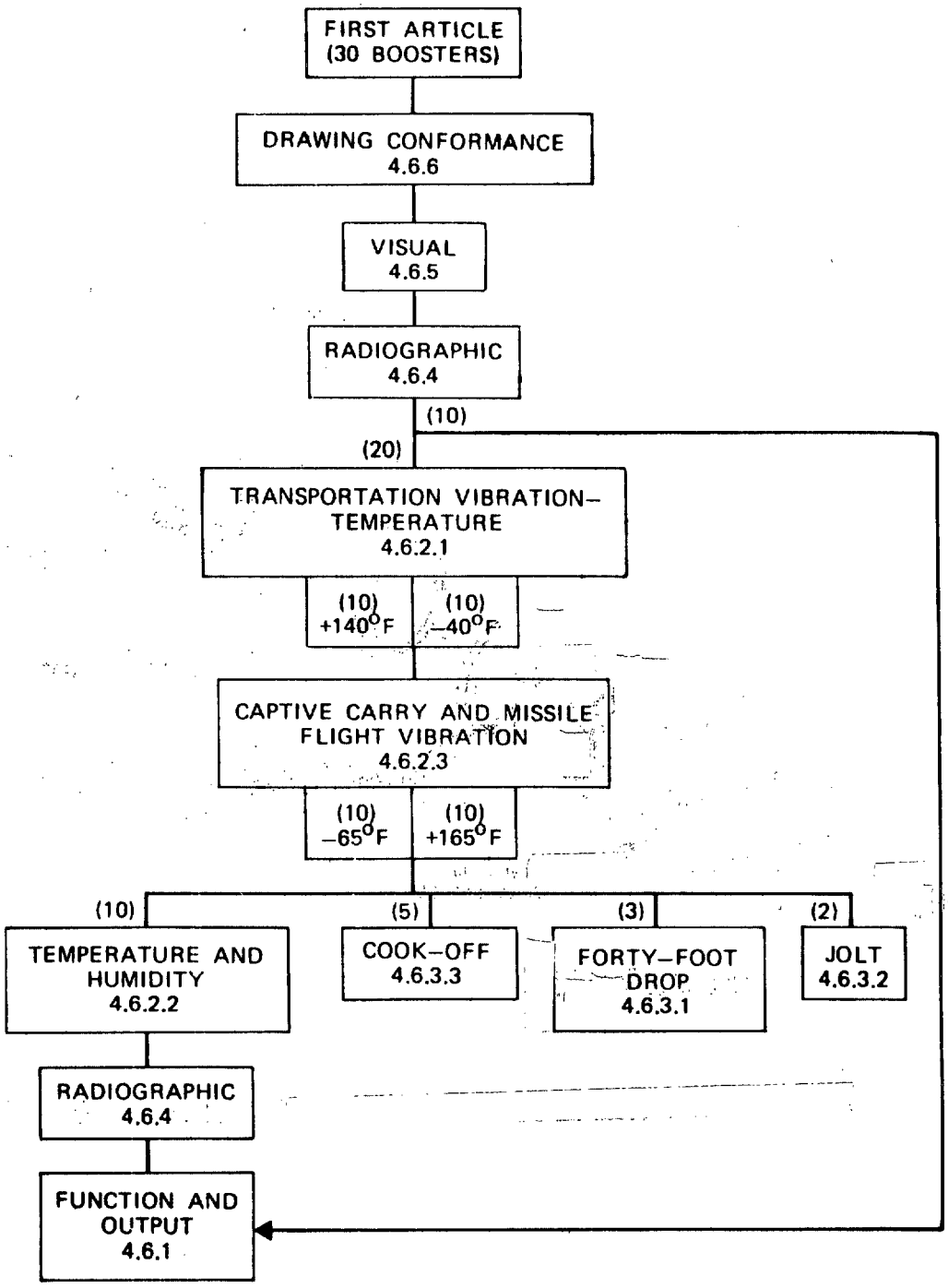


FIGURE 2. First article inspection sequence.

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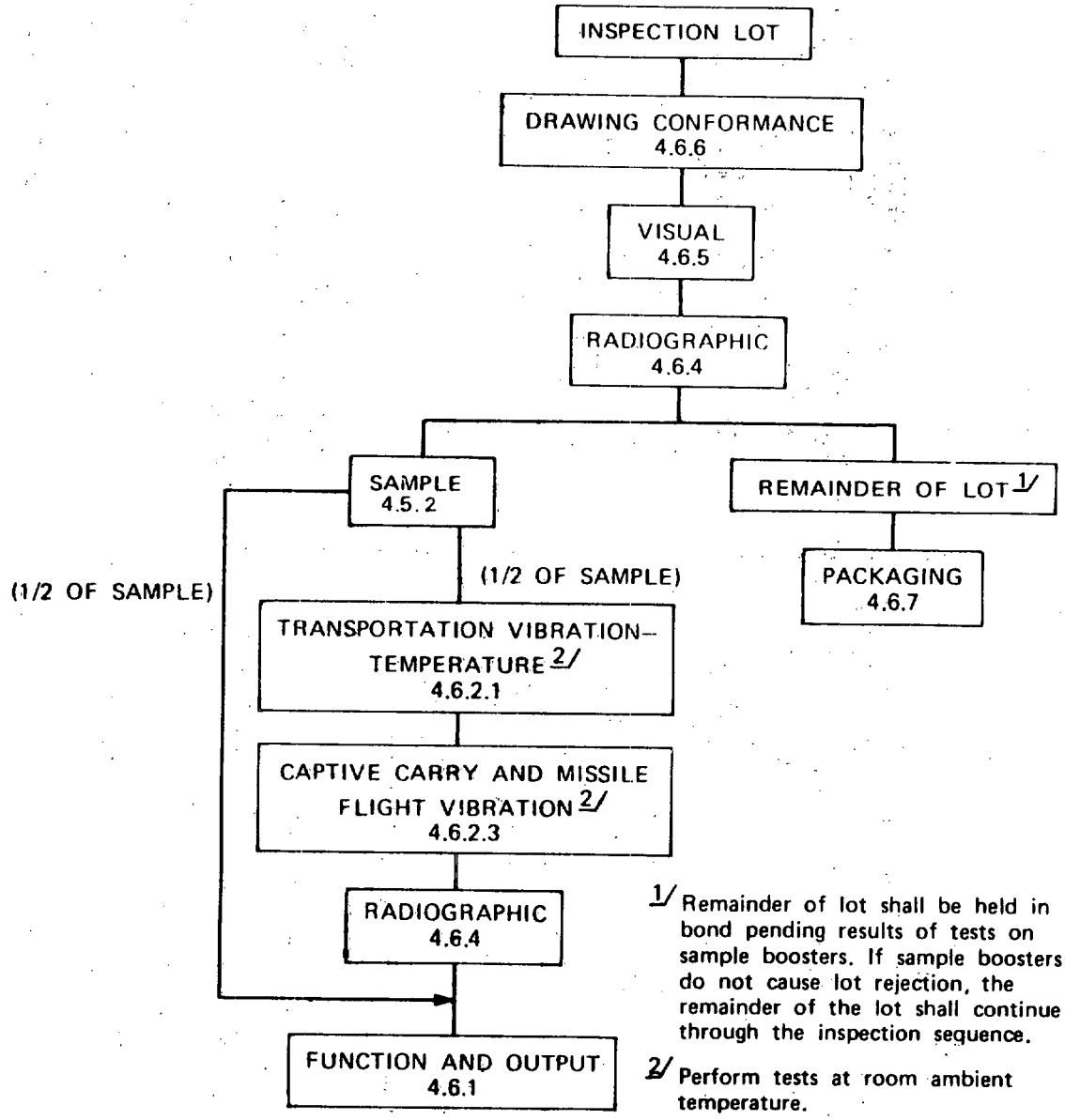
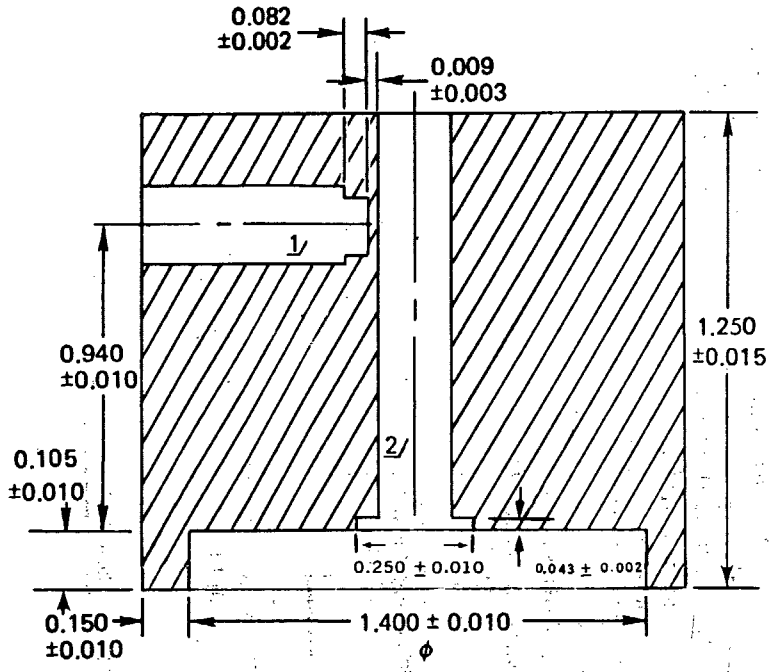


FIGURE 3. Inspection lot inspection sequence.

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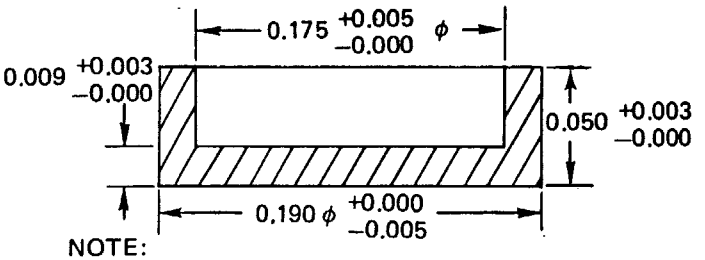
NOTES:

1. Material: Corrosion-resistant steel in accordance with ANSI/ASTM A-582, Type 416, Condition A. Heat-treat in accordance with MIL-H-6875 for Class B steels to Rockwell hardness C25 to 35.

1/ 0.175 +0.005, -0.000 ϕ to depth leaving 0.009 ±0.003 barrier; counterbore 0.193 +0.002, -0.000 ϕ to leave 0.082 ±0.002 shoulder.

2/ 0.193 +0.002, -0.000 ϕ through; counterbore 0.250 ±0.010 ϕ to a depth of 0.043 ±0.002.

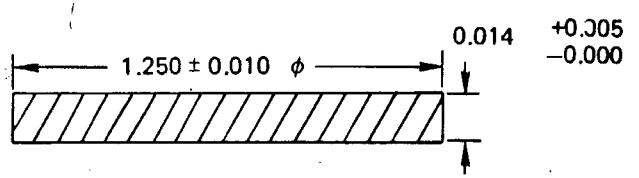
FIGURE 4a. Lead holder.



NOTE:

1. Material: Corrosion-resistant steel in accordance with ANSI/ASTM A-581 or A-582, Type 416, Condition A. Heat-treat in accordance with MIL-H-6875 for Class B steels to Rockwell hardness C25 to 35.

FIGURE 4b. Spacer.

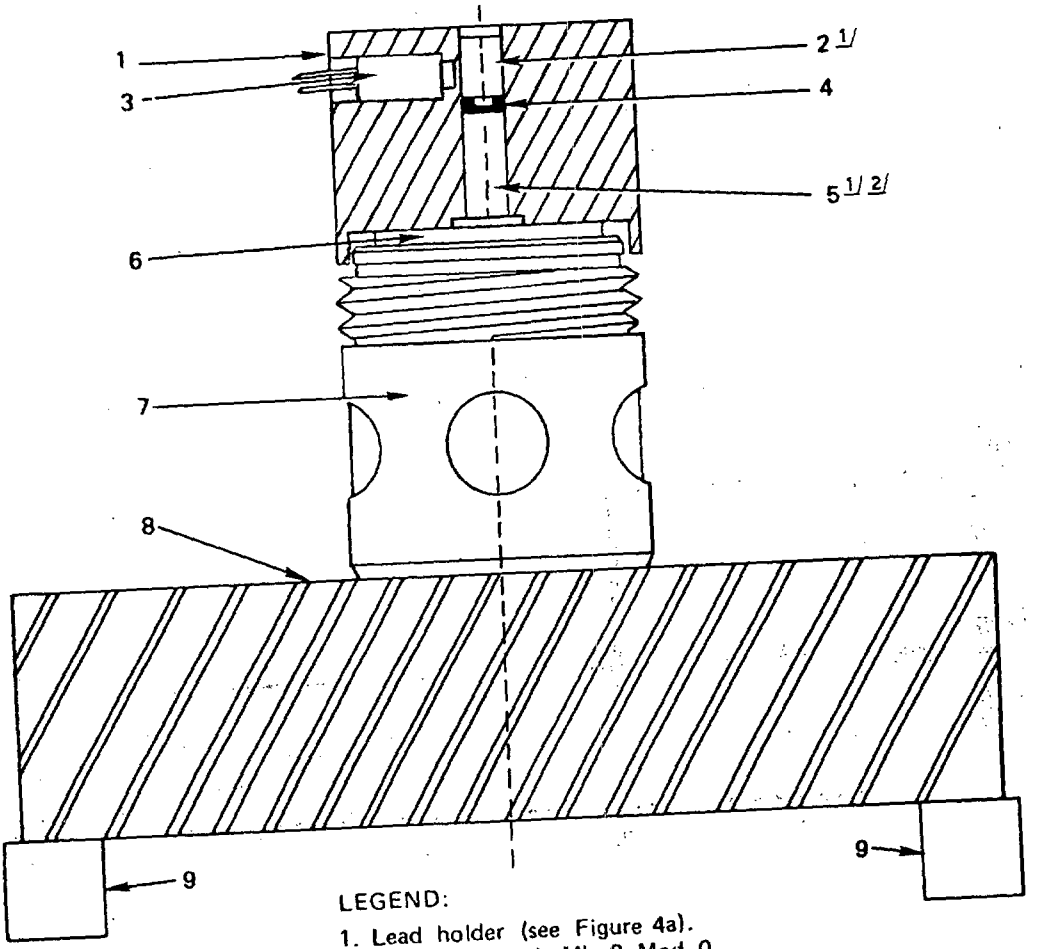


NOTES:

1. Material: Aluminum alloy, 6061, in accordance with QQ-A-225/8, Temper T6.
2. Figures 4a, 4b, and 4c: All dimensions are in inches; figures not to scale.

FIGURE 4c. Aluminum barrier.

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LEGEND:

1. Lead holder (see Figure 4a).
2. Explosive Lead, Mk 8 Mod 0.
3. Detonator.
4. Spacer (see Figure 4b).
5. Explosive Lead, Mk 16 Mod 0.
6. Aluminum barrier (see Figure 4c).
7. Fuze Booster, Mk 38 Mod 2.
8. Dent block (see 4.6.1.1.1.e).
9. Support blocks (see 4.6.1.1.1.f).

NOTES:

1/ Position Items 2 and 5 in Item 1, output (green) ends down.

2/ Secure Item 5 in Item 1 using a suitable adhesive. Type of adhesive used shall be at the contractor's discretion. There shall be a 0.043 ±0.002 air gap between Items 5 and 6 (see Figure 4a).

1. Figure not to scale.

FIGURE 5. Function and output test setup.

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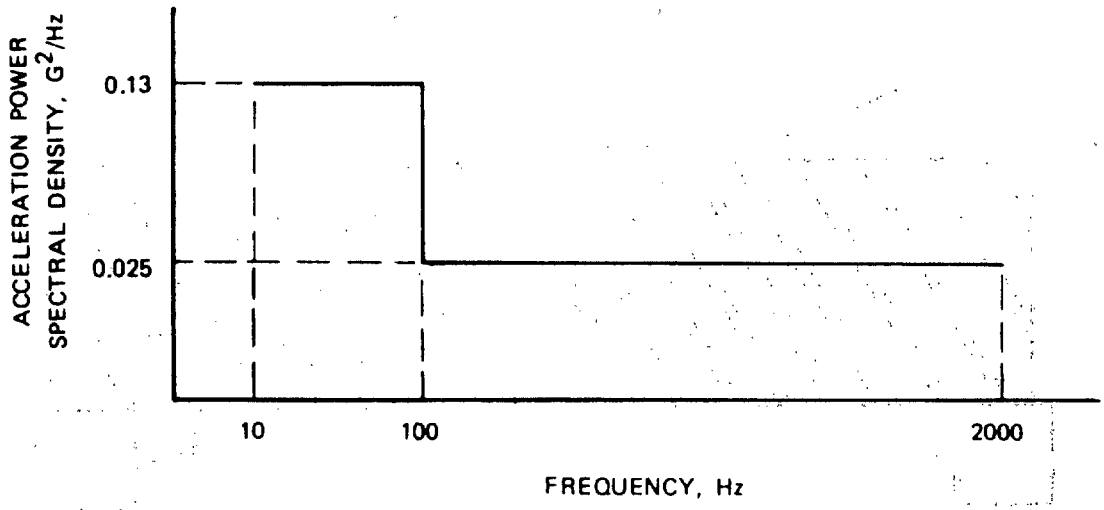
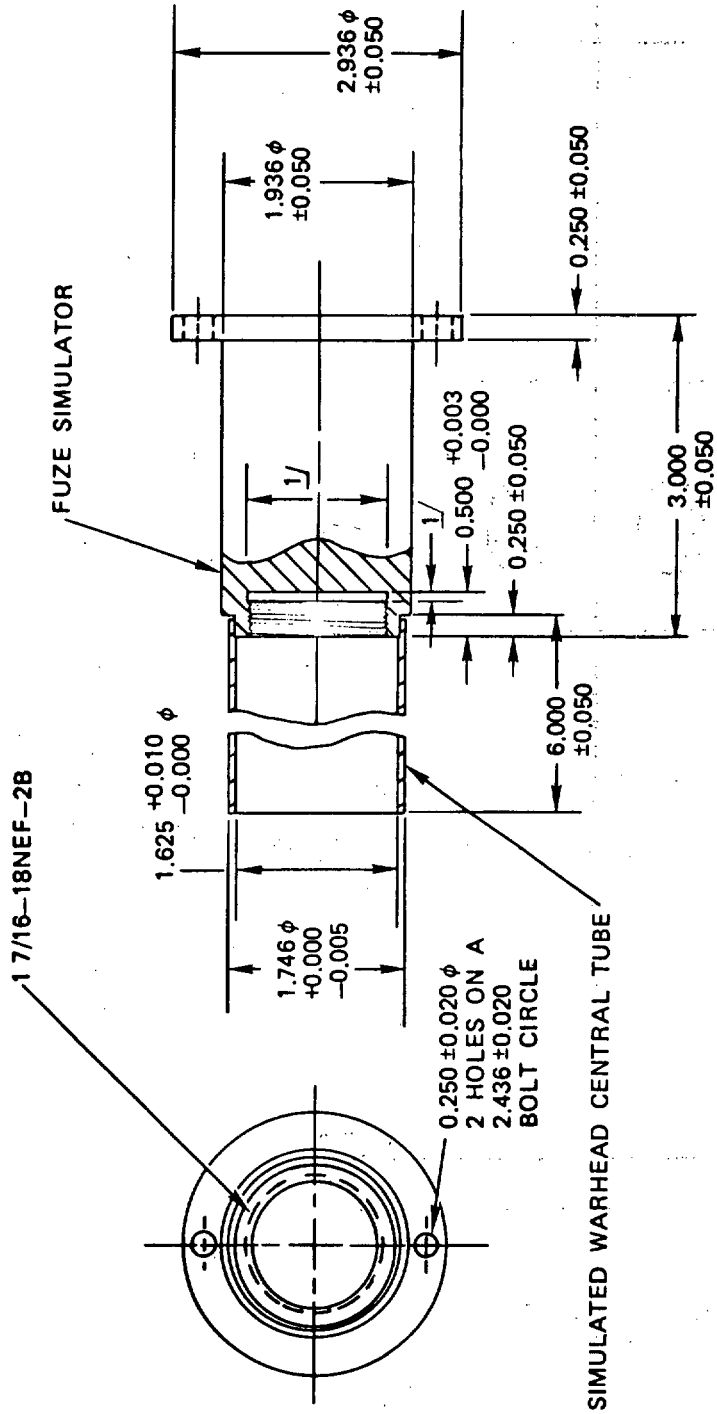


FIGURE 6. Power spectral density curve for captive carry and missile flight vibration test.

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NOTES:

1. All dimensions are in inches.
2. Materials:
   a. Simulated warhead central tube: Seamless carbon steel tubing in accordance with ANSI/ASTM A–519, grade and condition optional.
   b. Fuze simulator: Steel bar in accordance with ANSI/ASTM A–108, Grade 1117, 1118, or 1213.

1/ Thread relief: 0.094 +0.015, –0.000 X 1.445 diameter maximum.

FIGURE 7. Cook-off test fixture.

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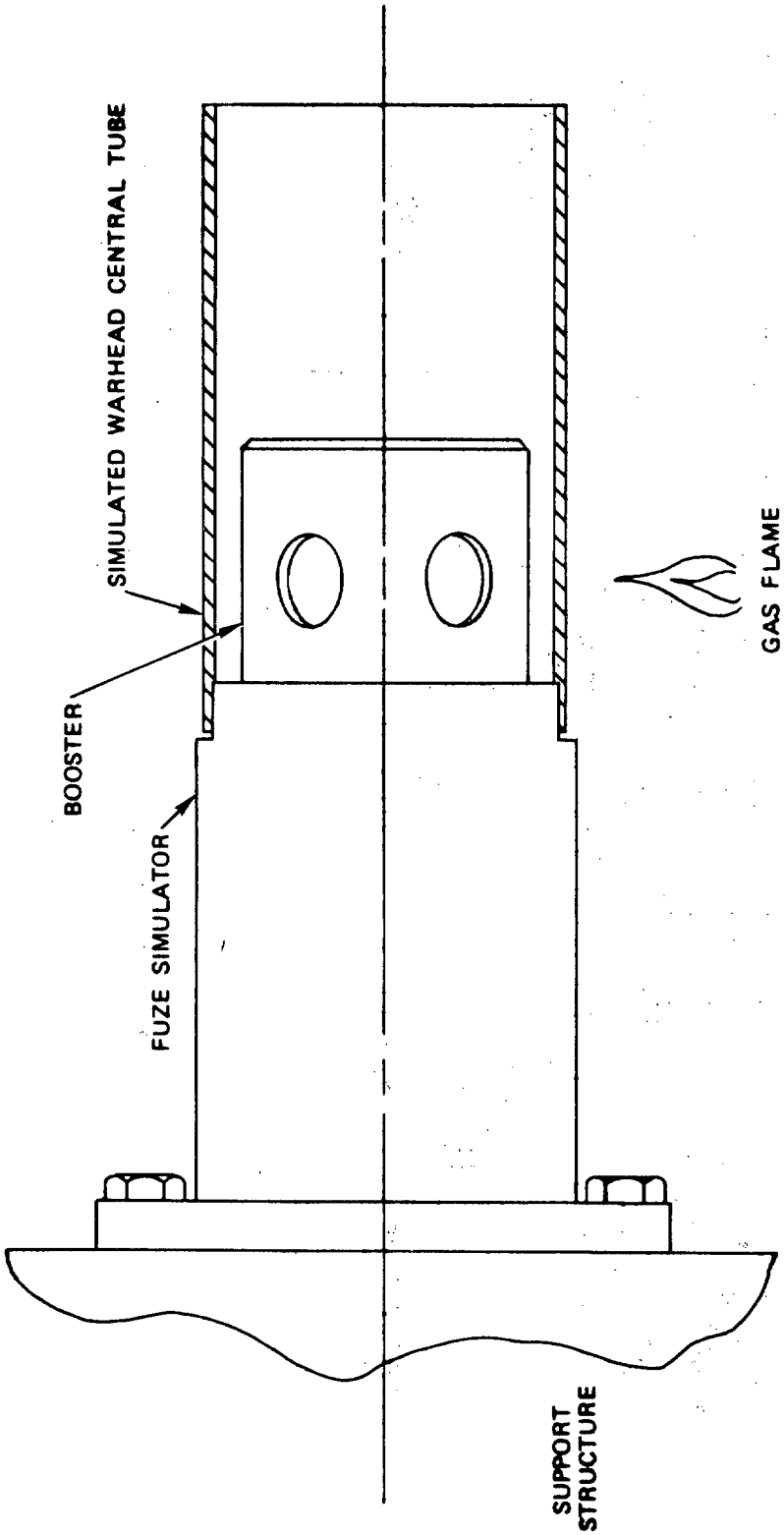
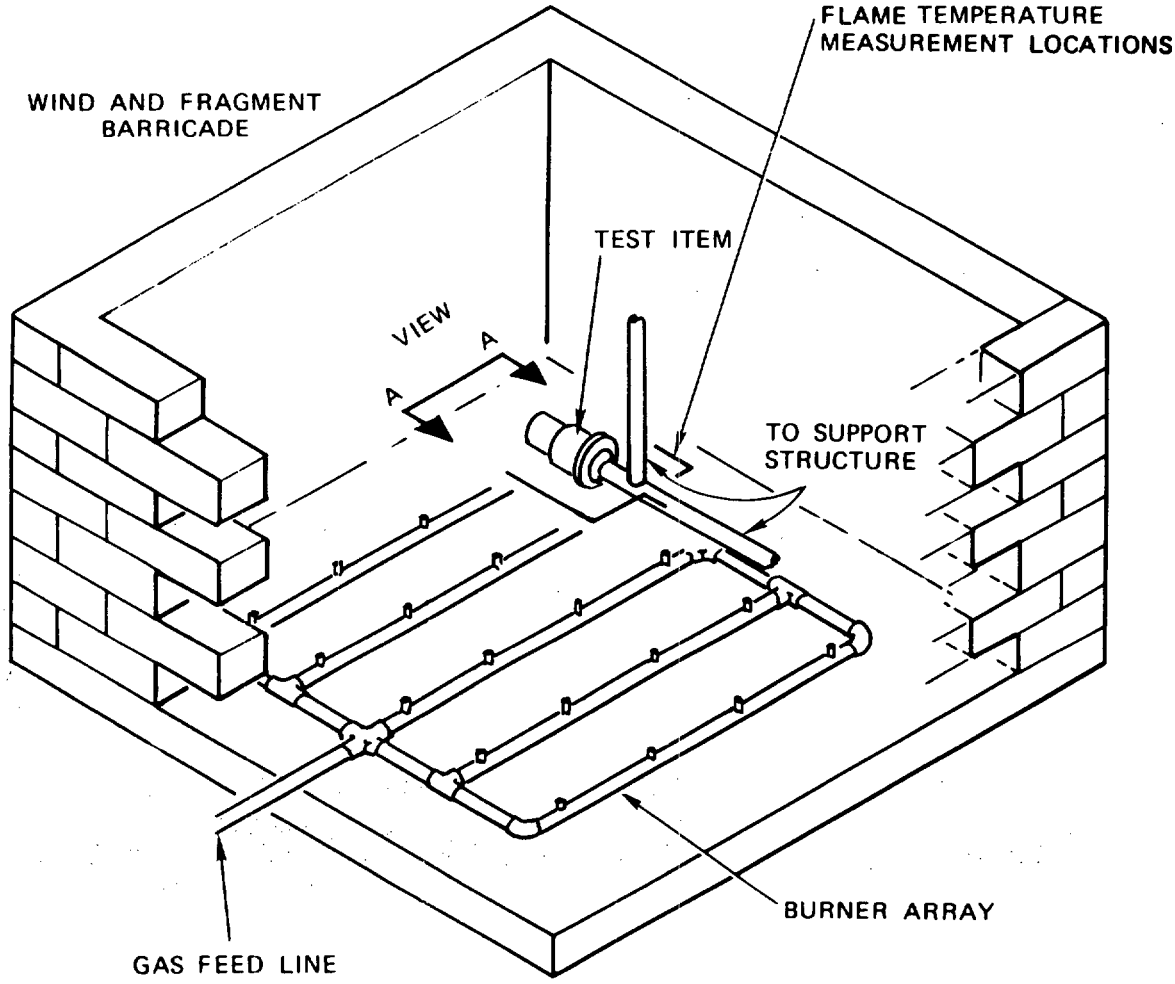


FIGURE 8. Booster installed in test fixture for cook-off test.

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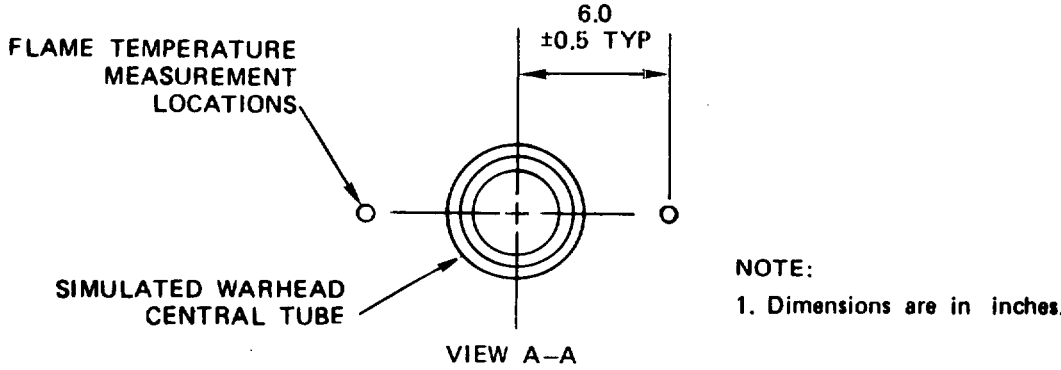


FIGURE 9. Cook-off test setup.

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